Ian Benedict Rego

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github.com/ianbenedictrego | Valid Student Visa | Eligible for Werkstudent and Internship Employment

PROFESSIONAL EXPERIENCE

Machine Learning Teaching Assistant, *Otto von Guericke Universität (OVGU)* — 10/2026 – Present, Magdeburg, Germany
- Delivered weekly tutorials on machine learning algorithms including regression, decision trees, neural networks, clustering, reinforcement learning, and model evaluation, emphasizing both technical rigor and intuitive understanding
- Guided students in implementing ML models in Python and analyzing experimental results

Research Project, *Autonomous Mutisensor System Labs* — 10/2026 – Present, Magdeburg, Germany
Project: **Learning Multiple Interacting Motion Models to Predict Maritime Vessel Trajectories**
- Developing a JAX-based maritime vessel tracking system using a six-state motion model and Kalman filtering, with AIS trajectory data to learn vessel-specific motion parameters, improve trajectory prediction, and explore Interacting Multiple Model (IMM) filtering.

Associate QA Engineer, *Sclera* — 03/2024 – 09/2025, Mangaluru, India
- Executed functional, regression, smoke and integration testing for web applications, identifying defects, validating fixes and supporting reliable feature releases
- Designed and executed REST API tests in a microservices environment, validating inter-service communication and collaborating with developers to investigate and resolve integration defects
- Supported Selenium-based test automation to expand regression coverage across key user journeys and reduce repetitive manual validation

QA Intern, *Sclera* — 01/2024 – 03/2024, Mangaluru, India
- Assisted with UI and API testing across asset-management workflows, including test-case design, regression testing and defect tracking.
- Collaborated with QA and development teams to understand system architecture and microservices-based communication

EDUCATION

M.Sc. in Digital Engineering (Data Science), *Otto von Guericke Universität (OVGU)* — 10/2025 – Present, Magdeburg, Germany

B.Eng. in Electrical & Electronics, *N.M.A.M. Institute of Technology* — 12/2020 – 05/2024, Karkala, India

TECHNICAL SKILLS

Programming: Python, Java | **Generative AI:** LangChain, LangGraph, Vector Databases, RAG, FAISS | **Deep Learning:** PyTorch | **Machine Learning & Data Science:** scikit-learn, NumPy, pandas, Matplotlib | **Evaluation:** LLM-as-a-Judge, LangSmith | **Backend:** FastAPI, Pydantic, REST APIs, Redis, SQL | **Software Testing:** Selenium, TestNG | **Cloud & DevOps:** AWS, Docker, Linux | **Version Control:** Git | **Agile Methodologies:** Scrum

LANGUAGES

English: C1 – CEFR | **German:** A2 – CEFR (actively progressing toward B1)

PROJECTS

Agentic LLM Router
- Developed an agentic GenAI routing service using LangGraph to dynamically route queries across Llama and Gemini models based on user intent and query complexity
- Designed a modular workflow with structured outputs, conditional routing, model fallbacks and LLM-as-a-Judge evaluation to assess routing and response quality
- Integrated Redis-based TTL caching and deployed the pipeline as a FastAPI service with automatically generated Swagger documentation, reducing redundant LLM calls

Enterprise GenAI Knowledge Retrieval Platform
- Built a hybrid enterprise RAG pipeline combining dense retrieval, BM25 and Reciprocal Rank Fusion for multi-document knowledge retrieval
- Integrated cross-encoder reranking and metadata-aware indexing to improve retrieval relevance and answer grounding
- Developed automated PDF ingestion, semantic chunking, embedding generation and persistent FAISS indexing for an end-to-end knowledge platform

RoadLens-Spatiotemporal Analysis of Collision Severity and Persistent Hotspots

- Developing a data science pipeline to integrate and preprocess UK road collision, vehicle, and casualty datasets for spatiotemporal analysis of collision severity and persistent accident hotspots.
- Designing machine learning models to classify collision severity and forecast future high-risk locations, evaluating the added predictive value of vehicle and casualty characteristics.
- Conducting geospatial hotspot analysis and road-user injury prioritisation to identify persistent high-severity locations and compare safety priorities for pedestrians, cyclists, and motorcyclists.

Bias Detection and Fairness Auditing Platform

- Developed an NLP/Generative AI pipeline for demographic-bias detection and LLM-assisted content rewriting.
- Evaluated subgroup-level bias and semantic similarity, achieving up to 18% fairness improvement while preserving 80% semantic similarity.
- Delivered results through automated reports and interactive Streamlit visualizations.

EXTRA CURRICULAR

- Member of the Academic Club at SI@FIN
- Selected for the Sakura Science Exchange program organised by JST and represented the EEE branch and the college at Ritsumeikan University, Japan.
- Member of Google Developer Student Club